

Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters

Abstract – this project focuses on analyzing sales trends and time based performance patterns for Afficionado coffee Roasters using Python based data analytics techniques. The study examines sales performance, including hours days of the week, and monthly trends to identify big business periods and customer purchasing behaviors. Data preprocessing application and visualization techniques were applied using Pandas Matplotlib, Seaborn and Plotly to generate meaningful insights from transaction sales data set the analysis revealed significant variations in sales performance across different time intervals helping identify high demand periods and revenue generating opportunities an interactive streamlined dashboard was also developed to provide dynamic exploration of sales trends and support driven decision making the findings can assist business stakeholders in optimizing staffing, inventory management, promotional activities and operational planning.

Keywords – sales train analysis, time based analysis, business intelligence, data analytics, Python, streamlined, data visualization, coffee shop analysis, revenue analysis

INTRODUCTION

In specially coffee retail businesses, understanding when sales occur is as important as understanding what products are sold. Customer demand often varies significantly across different hours of the day, creating challenges for staffing, inventory management, and operational planning. Poor understanding of temporal demand patterns can result in overstaffing during low-demand periods, understaffing during peak hours, inconsistent customer experiences, and increased operational costs.

Many retail managers rely on assumptions and experience when making operational decisions. However, data-driven analysis provides more reliable insights into customer behavior and sales performance. By analyzing transaction-level sales data, businesses can identify peak demand periods, optimize workforce allocation, improve customer service, and increase operational efficiency.

OBJECTIVE

Primary Objectives

- Identify overall sales patterns within the available 2025 transaction dataset.
- Determine peak and low-demand business hours.
- Analyze hourly revenue behavior and customer demand trends.

Secondary Objectives

- Compare temporal demand patterns across different store locations.
- Support staff scheduling and operational planning.
- Provide evidence-based explanations for observed sales behavior.
- Demonstrate the application of forecasting techniques for future revenue estimation.

DATA COLLECTION

The sales data for this study was collected from 3 store locations operating throughout 2025. The dataset includes transaction timestamps, transaction quantity and unit price and store identifiers. Data was sourced from the company's point of sale system logs verified for accuracy and completeness. Transactions were aggregated by time, and store location to support temporal and spatial sales trend analysis.

METHODOLOGY

Data Ingestion and Validation

The transaction dataset was imported into the python environment using the Pandas library.

```
# Data Loading and Cleaning
df=pd.read_csv("Afficionado Coffee
Roasters.csv")
```

Initially data inspection was performed to verify column names, data types, and overall dataset structure.

Data validation procedures were conducted to ensure data consistency.

```
df.info()
```

The Data validation process included checking for missing values, identifying duplicate records, and verifying logical consistency in transaction quantities and unit price.

```
# Removing Duplicates & Missing values
df.drop_duplicates(inplace=True)
df.dropna(inplace=True)
```

Data Preprocessing

To prepare the dataset for analysis, the transaction time field was converted into date-time format using Python. This enabled the extraction of time-based features required for temporal analysis.

```
# Defining Transaction time
df["transaction_time"] = pd.to_datetime(df["transaction_time"], format="%H%M%S %p")
```

Revenue for each transaction was calculated using following formula:

```
# Calculating Revenue
Revenue = transaction_qty * unit_price
```

The calculated revenue variable served as a key performance indicator throughout the study.

Feature Engineering

Several analytical features were derived from the original dataset to facilitate time-based analysis.

Hour Extraction

The transaction time variable was transformed to extract the hour component, allowing transactions to be grouped according to the time of occurrence.

```
# Hour of day
df["transaction_time"] = pd.to_datetime(df["transaction_time"])

df["hour"] = df["transaction_time"].dt.hour
```

Time Bucket Creation

Business hours were categorized into operational time segments:

- Morning: 6AM -11 AM
- Afternoon: 12PM-4PM
- Evening: 5PM-9PM

```
# Time buckets
def get_time_bucket(hour):
    if 6<=hour<=11:
        return "Morning"
    elif 12<=hour<=16:
        return "Afternoon"
    elif 17<=hour<=21:
        return "Evening"
    else:
        return "Late hours"

df['time_bucket'] = df["hour"].apply(get_time_bucket)
```

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify temporal sales patterns and revenue behavior.

- The analysis included:
- Hourly revenue distribution
- Peak-hour identification
- Revenue comparison across operational periods
- Store-level revenue comparison
- Store-wise hourly demand analysis

Visualizations were created using Matplotlib to support interpretation of the results and identify significant business patterns.

```
# Store-level Revenue comparison
store_revenue = df.groupby('store_location')['revenue'].sum().sort_values(ascending=False)
```

Store-level Temporal Analysis

Revenue generated by each store location was aggregated and compared across business hours. Hourly revenue curves were developed to examine similarities and differences in customer demand patterns among Astoria, Hell's kitchen, and Lower Manhattan locations. This analysis provided insights into location-specific operational requirements and staffing considerations.

```
# Store level temporal Demand
comparison
store_hourly=df.groupby(['store_location', 'hour'])['revenue'].sum().reset_index()
```

```
for store in
store_hourly['store_location'].unique(
):
    data=store_hourly[store_hourly['store_location']==store]
```

Regression Analysis

Linear Regression was applied to evaluate the relationship between business hours and revenue generation.

```
# Regression
from sklearn.linear_model import
LinearRegression
import numpy as np

X=hourly_revenue.index.values.reshape(-1,1)
y=hourly_revenue.values

model=LinearRegression()
model.fit(X,y)

print("R2 Score:", model.score(X,y))
```

The objective of this analysis was to determine whether revenue followed a measurable trend throughout the operating day. A regression line was fitted to hourly revenue values, and the coefficient of determination (R^2) was calculated to assess the exploratory power of the model.

ARIMA Forecasting

An autoregressive Integrated Moving Average (ARIMA) model was implemented to demonstrate forecasting capability using historical hourly revenue data.

The ARIMA model was trained using aggregated hourly revenue observations and used to generate short term revenue forecasts. Although the dataset lacked transaction date information required for long-term forecasting, ARIMA provided a framework for predictive analysis based on available temporal data.

```
# ARIMA model implementation for
hourly revenue forecasting
from statsmodels.tsa.arima.model
import ARIMA
```

```
model
=ARIMA(hourly_revenue,order=(1,1,1))
model_fit=model.fit()
```

Analytical Tools

The following technologies and libraries were used throughout the project:

- Python
- Jupyter Notebook
- Pandas
- Numpy
- Matplotlib
- Scikit-learn
- Statsmodels
- Streamlit

These tools supported data preparation, analysis, visualization, statistical modeling, and forecasting activities.

Interactive Dashboard Development

To operate the analytical framework and interactive web-based dashboard was developed using streamlined. The dashboard serves as the primary interface for visualizing and exploring time based sales patterns identified in the data set. It enables dynamic filtering of data across store locations time ranges and product categories allowing users to interactively analyze sales performance.

Application integrates multiple analytical modules including overall sales trend visualization, hourly demand distribution, and cross location analysis. Key performance indicators such as revenue and transaction volume are dynamically updated based on user-selected filters. Time-based insights such as morning rush hours, off-peak periods, and evening demand spikes are visualized using line charts, bar plots, and heatmaps.

Streamlit was chosen due to its lightweight architecture and ability to rapidly convert Python-based analytical models into interactive web applications without requiring separate front end development. This makes it particularly suitable for data driven decision support systems where real-time exploration of insights is required.

The final dashboard acts as a decision-support tool for retail managers, enabling evidence-based staffing optimization and operational planning.

The dashboard bridges the gap between data analysis and business decision-making by translating raw transactional data into actionable insights.

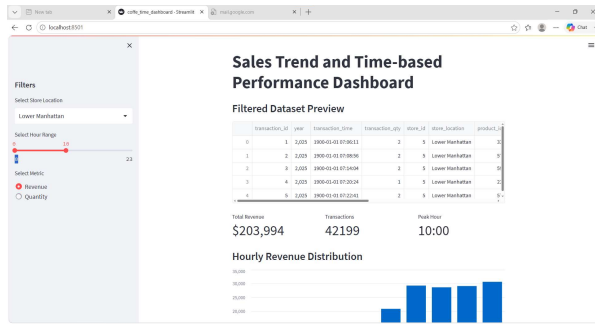


FIGURE 1: STREAMLIT DASHBOARD HOMEPAGE

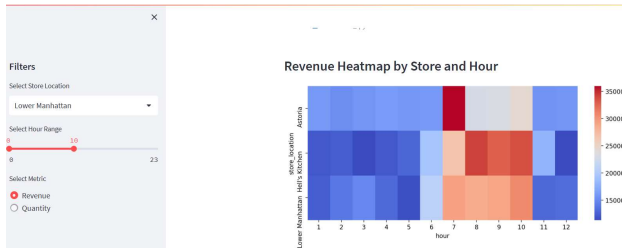


FIGURE 2: INTERACTIVE ANALYTICAL VISUALIZATIONS AVAILABLE WITHIN THE STREAMLIT DASHBOARD

ANALYSIS & RESULTS

A. Hourly Revenue Analysis

We analyzed hourly revenue patterns to identify peak and off peak sales period. Sales data was aggregated by each hour of the day across all store locations for the year 2025. The hourly revenue trends reveal specific time windows with consistently high sales, providing insights into customer purchasing behavior over the course of a day. These findings help optimize staffing and operational planning to match customer demand.

HOURLY REVENUE PATTERNS ACROSS ALL STORE LOCATIONS IN 2025

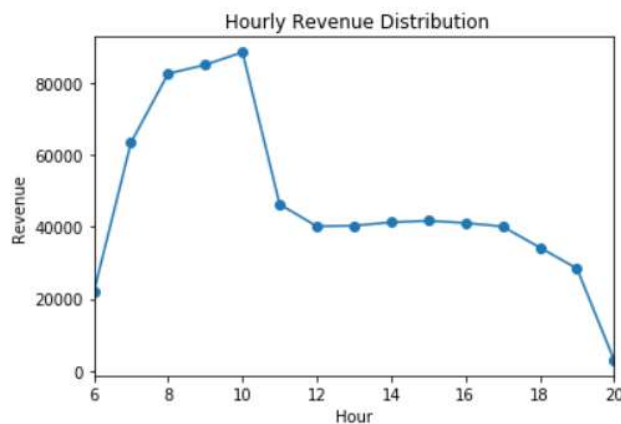


FIGURE 1: REVENUE VARIATION BY HOUR THROUGHOUT THE DAY, IDENTIFYING PEAK AND OFF-PEAK SALES PERIODS TO INFORM STAFFING AND OPERATIONAL DECISIONS

B. Peak Hour Identification

Peak hour identification involves aggregating sales data into hourly intervals and calculating the total revenue for each hour. By comparing these hourly totals, the hours with the highest revenue are designated as peak hours. To ensure reliability, this pattern is analyzed across year to confirm consistency. Identifying peak hours allows business to optimize resource allocation, including staffing, inventory management, to better meet customer demand during high-traffic periods.

```
# Peak hour Analysis using hourly transaction
hourly_transactions.idxmax()
```

C. Time Based demand Analysis

Morning rush hours were observed between 8:00 AM and 10:00 AM, with peak revenue generated at 10:00 AM. This indicates strong customer demand during the early business hours.

A noticeable decline in revenue was observed during the midday period (12:00 PM- 3:00 PM), indicating reduced customer activity compared to morning peak hours. Revenue gradually decreased during evening business hours. While customer activity continued after 5:00 PM to 8:00 PM, indicating that the majority of sales occurred during the morning hours.

TIME-BASED DEMAND ANALYSIS SHOWING REVENUE VARIATION BY HOUR

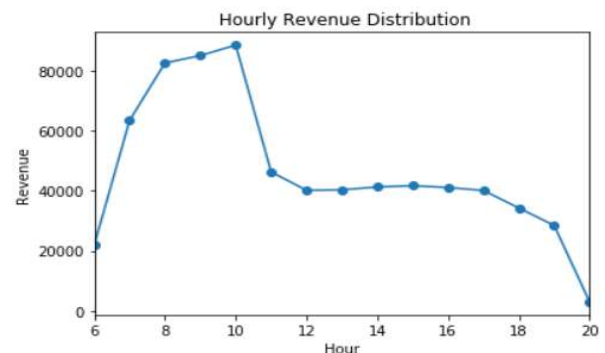


FIGURE 2: THE CHART ILLUSTRATES HOURLY REVENUE FLUCTUATIONS THROUGHOUT THE DAY, HIGHLIGHTING PEAK DEMAND PERIODS

D. Cross-level Temporal Comparison

Store-wise Line Graph

Store level temporal comparison revealed variations in hourly revenue patterns across locations. While all stores experienced higher demand during morning business hours, the timing and magnitude of peak revenue differed

slightly. Understanding these findings suggest that to allocate staff and stock efficiently, maximizing store performance and customer satisfaction.



FIGURE 3: COMPARISON OF HOURLY REVENUE ACROSS MULTIPLE STORE LOCATIONS

Heatmap

The heat map provides a comparative visualization of revenue generated across different store locations throughout the operating hours. Darker color intensities indicate higher revenue levels, enabling quick identification of peak demand periods for each store. The results show that all three store locations experienced higher revenue during morning hours, particularly between 8:00AM and 10:00 AM. Although the overall demand pattern remained similar across locations, variations in revenue intensity suggest differences in customer traffic and purchasing behavior. These findings support location-specific staffing and operational planning rather than applying a uniform strategy across all stores.

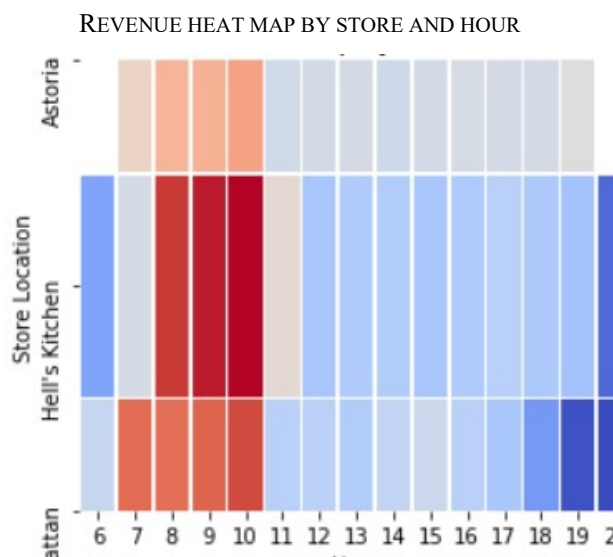


FIGURE 4: HOURLY REVENUE HEATMAP ACROSS STORE LOCATIONS.

E. Regression Analysis

Regression analysis interpretation involves understanding the relationship between a transaction hour and revenue generation. Revenue levels were highest during the morning period, particularly between 8:00 AM and 10:00 AM, with 10:00 AM identified as the peak revenue-generating hour. The downward trend line suggests that revenue generally decreases as the day progresses. The model achieved an R^2 score of 0.372, indicating a moderate relationship between transaction time and revenue.

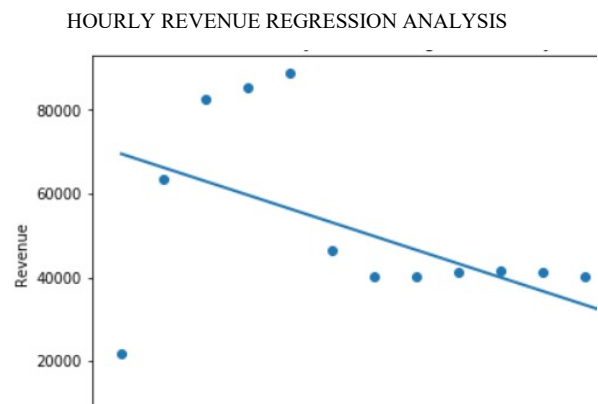


FIGURE 5 : LINEAR REGRESSION MODEL SHOWING THE RELATIONSHIP BETWEEN TRANSACTION HOUR AND REVENUE GENERATION

F. ARIMA Based Forecasting

An ARIMA (1, 1, 1) model was applied to the hourly revenue series to estimate future revenue patterns. The forecasting results provide an indication of expected revenue behavior beyond the observed transaction hours. Although the dataset contains a limited number of observations, the model demonstrates the application of time-series forecasting techniques for sales prediction and business planning.

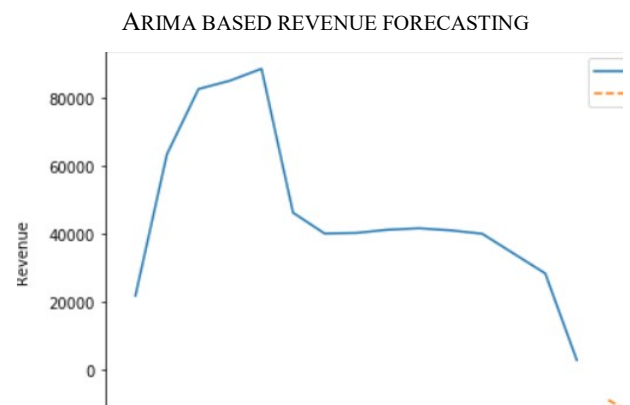
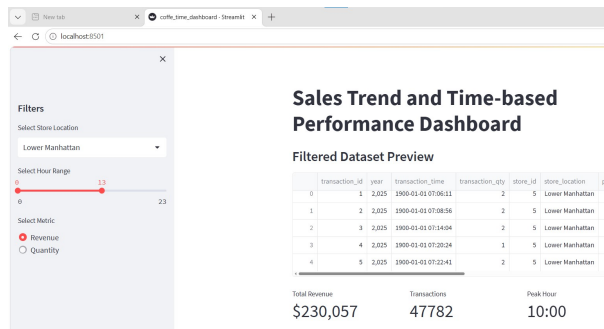


FIGURE 6: ARIMA –BASED FORECAST OF FUTURE REVENUE TRENDS USING HOURLY SALES DATA

G. Streamlit Dashboard



Streamlit dashboard integrates all analytical components into a unified interface, allowing real-time interaction, allowing real-time interaction with sales trends, temporal patterns, and location based insights.

DISCUSSION

The present study investigated the temporal sales performance of Afficionado Coffee Roasters using transaction-level retail data from the year 2025. The analysis focused on identifying hourly sales patterns, comparing store-level demand, and evaluating the applicability of statistical and forecasting techniques for operational decision-making.

The hourly revenue analysis revealed that customer demand is not uniformly distributed throughout the business day. Revenue gradually increased during the early morning hours and reached its highest value at 10:00 AM, making it the peak revenue-generating hour. This indicates that customers predominantly purchase coffee products during morning hours, which is consistent with typical consumer behavior where coffee is commonly consumed before or during work and educational activities. Following the morning peak, revenue gradually declined throughout the afternoon and evening, indicating comparatively lower customer demand during later business hours.

The time-based demand analysis further supported these observations. Morning hours consistently generated the highest revenue and transaction activity, demonstrating that this period represents the busiest operational window for the business. Afternoon hours showed moderate demand, while evening hours showed moderate demand, while evening hours experienced a noticeable decline in revenue. These findings suggest that staffing levels, inventory replenishment, and operational resources should be allocated according to hourly customer demand rather than maintaining a uniform distribution throughout the day.

Store-level temporal comparison demonstrated that all three store locations- Astoria, Hell's Kitchen, and Lower Manhattan – followed a broadly similar hourly demand pattern. Although the overall trend remained consistent

across location, differences were observed in revenue intensity, indicating variations in customer volume and purchasing behavior. Such differences emphasize the importance of implementing location-specific operational strategies instead of applying identical staffing and inventory policies across all stores.

The hourly revenue heatmap provided effective visual representation of temporal demand across store locations. Higher color intensity during morning business hours clearly highlighted peak demand periods, allowing management to quickly identify critical operational windows. The heatmap also facilitated easy comparison between store locations and supported evidence-based managerial decision-making.

Linear Regression analysis was performed to evaluate the relationship between business hours and revenue generation. The regression model indicated a measurable relationship between time and revenue, suggesting that business hours influence sales performance. Although revenue is affected by several external factors beyond time alone, the regression analysis successfully quantified the overall trend and provided additional statistical support for the observed hourly sales pattern.

To complement descriptive analysis, an ARIMA forecasting model was developed using aggregated hourly revenue data. The forecasting results demonstrated that historical hourly sales information can be effectively utilized to estimate future revenue patterns. Although the available dataset lacked transaction-date information required for long-term forecasting, the ARIMA model successfully illustrated the practical application of time-series forecasting for short-term operational planning. Such forecasting models can assist managers in anticipating customer demand, optimizing workforce allocation, and improving inventory planning.

One important limitation encountered during this study was the absence of a transaction-date column. Since only the transaction-year and transaction-time were available, analyses involving daily trends, weekly patterns, monthly seasonality, and day-of-week comparisons could not be performed. Consequently, the scope of this research was limited to hourly temporal analysis. Nevertheless, the available data proved sufficient for identifying meaningful operational insights related to customer demand across business hours.

Overall, the findings demonstrate that time-based sales analytics provides valuable support for operational decision-making within specialty coffee retail businesses. By combining exploratory data analysis, statistical modeling, forecasting techniques, and interactive dashboard visualization, the study offers a comprehensive framework for understanding customer demand patterns and improving business performance through evidence-based decisions.

RECOMMENDATIONS

- Increase staffing during morning peak hours.
- Maintain adequate inventory during morning rush.
- Review evening operating strategy due to lower demand.
- Use forecasting models for workforce and inventory planning.
- Monitor store-specific demand patterns instead of applying a uniform staffing strategy.

CONCLUSION

This project successfully analyzed the sales performance of Afficionado Coffee Roasters using time-based transaction data. The primary objective was to understand when sales occur and how demand varies across different hours of the day and store locations.

Through exploratory data analysis and visualization, clear temporal patterns were identified in customer purchasing behavior. The analysis revealed that sales are highly dependent on time of day, with peak revenue occurring during morning hours, followed by a gradual decline during afternoon and evening periods. This indicates strong time-sensitive customer demand behavior.

Store-wise comparison showed that each location follows a slightly different sales pattern, highlighting the importance of location-specific operational planning. Some stores experienced higher peak-hour intensity, while others showed more balanced distribution across the day.

Feature engineering enabled the extraction of meaningful time-based variables such as hour of day and time buckets, which helped in deeper analysis of demand patterns. Regression analysis further helped in understanding the relationship between time and revenue trends, while ARIMA forecasting provided insights into future sales behavior.

The findings from this project can be directly applied to business operations. They support better staff scheduling, optimized store management, and improved resource allocation during peak and off-peak hours. Additionally, the Streamlit dashboard provides an interactive platform for real-time exploration of sales trends.

Overall, this study demonstrates how data-driven techniques can transform raw transaction data into actionable business insights, enabling more efficient and informed decision-making in retail operations.

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